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LAB REPORT ON ISP NETWORKING DESIGHN

(OSPF, VLANs, DHCP, DNS & INTERNET CONNCETIVITY

Objectives:

- To design and implement a 3-router ISP network topology in Cisco Packet Tracer.
- To configure OSPF (Open Shortest Path First) dynamic routing on all routers.
- To implement VLANs for customer network segmentation on Layer 2 switches.
- To configure DHCP pools for automatic IP assignment across multiple VLANs.
- To set up DNS server for domain name resolution within the ISP network.
- To apply supernetting/route aggregation to reduce routing table size.
- To configure a default route and propagate internet connectivity via OSPF.
- To verify and test end-to-end connectivity across all customer sites.

Software Requirement:

- Cisco Packet Tracer (version 8.0 or later recommended)
- Windows / Linux PC

Theory:

Network Overview

This lab simulates a small Internet Service Provider (ISP) network that serves two customer sites. The ISP core router (R1) connects to the internet and acts as the backbone. Customer Site A (R2) and Customer Site B (R3) are connected to R1 using point-to-point /30 links. Each customer site has three VLANs served by a Layer 2 switch, a dedicated DHCP server handles dynamic addressing for all VLANs, and a DNS server resolves hostnames to IP addresses across the network.

OSPF (Open Shortest Path First)

OSPF is a link-state interior gateway routing protocol (IGP) defined by RFC 2328. It uses Dijkstra's algorithm to build a shortest path tree and compute optimal routes. Unlike distance-vector protocols, OSPF converges quickly and scales well for enterprise and ISP deployments.

Advantages:

- Fast convergence and loop-free topology
- Supports VLSM and CIDR for efficient IP addressing
- Hierarchical design using areas reduces overhead
- Multicast-based updates (224.0.0.5 / 224.0.0.6) reduce bandwidth usage

Supernetting / Route Aggregation

Supernetting (also called route summarization or route aggregation) combines multiple contiguous subnets into a single, larger summary route. This reduces the number of entries in routing tables, decreasing memory usage and speeding up routing decisions. In this lab, the supernet 192.168.0.0/21 is used, covering 2048 addresses, subdivided into eight /24 subnets.

Advantages:

- Reduces routing table size across the ISP backbone
- Decreases OSPF LSA flooding overhead
- Masks internal topology changes from other routers

VLANs (Virtual Local Area Networks)

VLANs logically segment a physical switch into multiple broadcast domains. Each VLAN acts as an independent Layer 2 network, improving security and reducing unnecessary broadcast traffic. In this lab, Router-on-a-Stick (RoaS) is used: a single physical router interface carries traffic for multiple VLANs via 802.1Q sub-interfaces.

Advantages:

- Isolates customer traffic for security
- Reduces broadcast domain size, improving performance
- Enables flexible network segmentation without additional hardware
-

DHCP (Dynamic Host Configuration Protocol)

DHCP automates IP address assignment to client devices. A centralized DHCP server handles all six VLAN pools (three per customer). Routers forward DHCP broadcast requests to the server using the ip helper-address command (DHCP relay).

Advantages:

- Eliminates manual IP configuration on every device
- Prevents IP address conflicts
- Centralized management of all address pools

DNS (Domain Name System)

DNS translates human-readable hostnames (e.g., www.customerA.com) into IP addresses. A single DNS server maintains A records for customer sites and ISP infrastructure. DHCP automatically provides the DNS server IP to all clients.

Advantages:

- Allows access to resources using domain names instead of IP addresses
- Simplifies network management when IPs change
- Supports scalable naming for growing networks

IP Addressing & Supernetting Plan:

Supernet Design

The supernet 192.168.0.0/21 is used, providing:

- 2048 total IP addresses
- Can be subdivided into 8 × /24 networks
- Address range: 192.168.0.0 to 192.168.7.255

Complete IP Address Table

Network Segment	Network Address	Subnet Mask	Usable IPs
R1 ↔ R2 Link	192.168.0.0/30	255.255.255.252	2
R1 G0/0	192.168.0.1	-	-
R2 G0/0	192.168.0.2	-	-
R1 ↔ R3 Link	192.168.0.4/30	255.255.255.252	2
R1 G0/1	192.168.0.5	-	-
R3 G0/0	192.168.0.6	-	-
Customer A – VLAN 10	192.168.1.0/24	255.255.255.0	254
Customer A – VLAN 20	192.168.2.0/24	255.255.255.0	254
Customer A – VLAN 30	192.168.3.0/24	255.255.255.0	254
Customer B – VLAN 10	192.168.4.0/24	255.255.255.0	254
Customer B – VLAN 20	192.168.5.0/24	255.255.255.0	254
Customer B – VLAN 30	192.168.6.0/24	255.255.255.0	254
Management / Servers	192.168.7.0/24	255.255.255.0	254

Server IP Addresses

Device	IP Address	Subnet Mask	Default Gateway
DHCP Server	192.168.7.10	255.255.255.0	192.168.7.1
DNS Server	192.168.7.20	255.255.255.0	192.168.7.1

R1 G0/2 (Server GW)	192.168.7.1	255.255.255.0	-

Network Topology:

The ISP network consists of three Cisco 2911 routers, two Layer 2 switches (2960) for customer sites, one server switch (SW-Server), two servers (DHCP and DNS), six PCs (three per customer), and one Internet cloud.

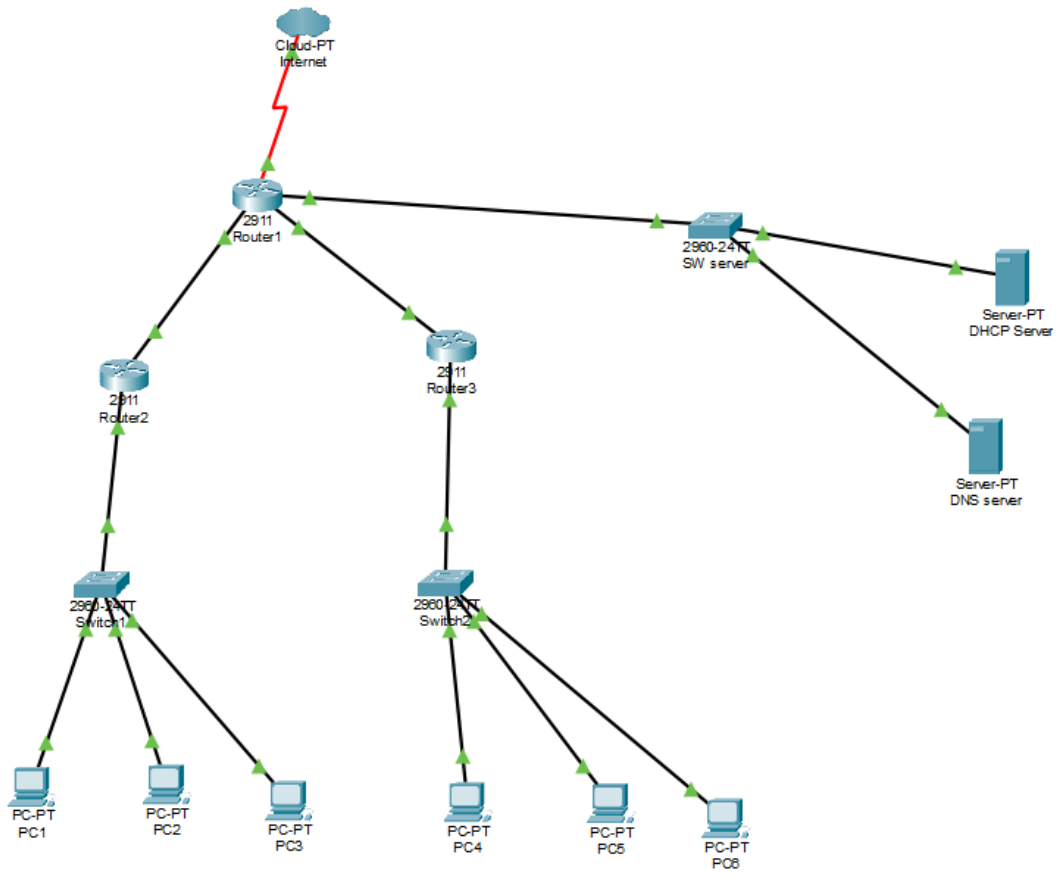


Fig: Network Topology

Topology Overview:

- R1 (Core ISP Router) – connects to R2, R3, server network, and the Internet.
- R2 (Customer Site A) – connected to R1 via G0/0; runs Router-on-a-Stick on G0/1 to SW1.
- R3 (Customer Site B) – connected to R1 via G0/0; runs Router-on-a-Stick on G0/1 to SW2.

- SW1 – Layer 2 switch for Customer A; trunks VLANs 10, 20, 30 to R2.
- SW2 – Layer 2 switch for Customer B; trunks VLANs 10, 20, 30 to R3.
- SW-Server – connects DHCP-Server and DNS-Server to R1 G0/2.
- Internet Cloud – WAN emulation device connected to R1 via Serial0/0/0.

Procedure:

Step 1: Topology Setup in Cisco Packet Tracer

1. Open Cisco Packet Tracer.
2. Add 3 × Cisco 2911 Routers and label them R1, R2, R3.
3. Add 3 × Cisco 2960 Switches and label them SW1, SW2, SW-Server.
4. Add 6 PCs: PC1–PC3 (Customer A), PC4–PC6 (Customer B).
5. Add 2 Servers: DHCP-Server and DNS-Server.
6. Add 1 Cloud-PT labeled Internet.

Step 2: Cabling Connections

From Device	Interface	To Device	Interface	Cable Type
R1	G0/0	R2	G0/0	Straight-Through
R1	G0/1	R3	G0/0	Straight-Through
R1	G0/2	SW-Server	G0/1	Straight-Through
R2	G0/1	SW1	G0/1	Straight-Through
R3	G0/1	SW2	G0/1	Straight-Through
R1	S0/0/0	Internet	Ethernet6	Serial/Straight
PC1	Fa0	SW1	Fa0/1	Straight-Through
PC2	Fa0	SW1	Fa0/2	Straight-Through
PC3	Fa0	SW1	Fa0/3	Straight-Through

PC4	Fa0	SW2	Fa0/1	Straight-Through
PC5	Fa0	SW2	Fa0/2	Straight-Through
PC6	Fa0	SW2	Fa0/3	Straight-Through
DHCP-Server	Fa0	SW-Server	Fa0/1	Straight-Through
DNS-Server	Fa0	SW-Server	Fa0/2	Straight-Through

Step 3: Router R1 Configuration (Core ISP Router)

```
Router>enable
Router#configure terminal
Router(config)#hostname R1
R1(config)#interface gigabitEthernet 0/0
R1(config-if)#description Link to R2
R1(config-if)#ip address 192.168.0.1 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface gigabitEthernet 0/1
R1(config-if)#description Link to R3
R1(config-if)#ip address 192.168.0.5 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface gigabitEthernet 0/2
R1(config-if)#description Server Network
R1(config-if)#ip address 192.168.7.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface serial 0/0/0
R1(config-if)#description Internet Connection
```

```
R1(config-if)#ip address 203.0.113.1 255.255.255.252
R1(config-if)#clock rate 64000
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#exit
R1#copy running-config startup-config
```

Step 4: Router R2 Configuration (Customer Site A)

```
Router>enable
Router#configure terminal
Router(config)#hostname R2
R2(config)#interface gigabitEthernet 0/0
R2(config-if)#description Link to R1
R2(config-if)#ip address 192.168.0.2 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface gigabitEthernet 0/1
R2(config-if)#description Trunk to SW1
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface gigabitEthernet 0/1.10
R2(config-if)#encapsulation dot1Q 10
R2(config-if)#ip address 192.168.1.1 255.255.255.0
R2(config-if)#exit
R2(config)#interface gigabitEthernet 0/1.20
R2(config-if)#encapsulation dot1Q 20
R2(config-if)#ip address 192.168.2.1 255.255.255.0
R2(config-if)#exit
R2(config)#interface gigabitEthernet 0/1.30
```

```
R2(config-if)#encapsulation dot1Q 30
R2(config-if)#ip address 192.168.3.1 255.255.255.0
R2(config-if)#exit
R2(config)#exit
R2#copy running-config startup-config
```

Step 5: Router R3 Configuration (Customer Site B)

```
Router>enable
Router#configure terminal
Router(config)#hostname R3
R3(config)#interface gigabitEthernet 0/0
R3(config-if)#description Link to R1
R3(config-if)#ip address 192.168.0.6 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#interface gigabitEthernet 0/1
R3(config-if)#description Trunk to SW2
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#interface gigabitEthernet 0/1.10
R3(config-if)#encapsulation dot1Q 10
R3(config-if)#ip address 192.168.4.1 255.255.255.0
R3(config-if)#exit
R3(config)#interface gigabitEthernet 0/1.20
R3(config-if)#encapsulation dot1Q 20
R3(config-if)#ip address 192.168.5.1 255.255.255.0
R3(config-if)#exit
R3(config)#interface gigabitEthernet 0/1.30
R3(config-if)#encapsulation dot1Q 30
```

```
R3(config-if)#ip address 192.168.6.1 255.255.255.0
R3(config-if)#exit
R3(config)#exit
R3#copy running-config startup-config
```

Step 6: OSPF Configuration on All Routers

On R1 (Core):

```
R1#configure terminal
R1(config)#router ospf 1
R1(config-router)#router-id 1.1.1.1
R1(config-router)#network 192.168.0.0 0.0.0.3 area 0
R1(config-router)#network 192.168.0.4 0.0.0.3 area 0
R1(config-router)#network 192.168.7.0 0.0.0.255 area 0
R1(config-router)#default-information originate
R1(config-router)#exit
```

On R2 (Customer A) – with route summarization:

```
R2(config)#router ospf 1
R2(config-router)#router-id 2.2.2.2
R2(config-router)#network 192.168.0.0 0.0.0.3 area 0
R2(config-router)#network 192.168.1.0 0.0.0.255 area 0
R2(config-router)#network 192.168.2.0 0.0.0.255 area 0
R2(config-router)#network 192.168.3.0 0.0.0.255 area 0
R2(config-router)#area 0 range 192.168.1.0 255.255.252.0
R2(config-router)#exit
```

On R3 (Customer B) – with route summarization:

```
R3(config)#router ospf 1
R3(config-router)#router-id 3.3.3.3
```

```
R3(config-router)#network 192.168.0.4 0.0.0.3 area 0
R3(config-router)#network 192.168.4.0 0.0.0.255 area 0
R3(config-router)#network 192.168.5.0 0.0.0.255 area 0
R3(config-router)#network 192.168.6.0 0.0.0.255 area 0
R3(config-router)#area 0 range 192.168.4.0 255.255.252.0
R3(config-router)#exit
```

Step 7: Default Route & Internet Connectivity

```
R1(config)#ip route 0.0.0.0 0.0.0.0 serial 0/0/0
  -- OR --
R1(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.2
R1(config)#router ospf 1
R1(config-router)#default-information originate
```

Step 8: VLAN Configuration on Switch SW1 (Customer A)

```
Switch>enable
Switch#configure terminal
Switch(config)#hostname SW1
SW1(config)#vlan 10
SW1(config-vlan)#name CustomerA_Dept1
SW1(config-vlan)#exit
SW1(config)#vlan 20
SW1(config-vlan)#name CustomerA_Dept2
SW1(config-vlan)#exit
SW1(config)#vlan 30
SW1(config-vlan)#name CustomerA_Dept3
SW1(config-vlan)#exit
SW1(config)#interface gigabitEthernet 0/1
SW1(config-if)#switchport mode trunk
```

```
SW1(config-if)#switchport trunk allowed vlan 10,20,30
SW1(config-if)#exit
SW1(config)#interface fastEthernet 0/1
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 10
SW1(config-if)#exit
SW1(config)#interface fastEthernet 0/2
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 20
SW1(config-if)#exit
SW1(config)#interface fastEthernet 0/3
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 30
SW1(config-if)#exit
SW1#copy running-config startup-config
```

Step 9: DHCP Server Configuration

Assign static IP to DHCP-Server: IP 192.168.7.10, Subnet 255.255.255.0, Gateway 192.168.7.1, DNS 192.168.7.20.

DHCP Pools:

Pool Name	Start IP	Gateway	DNS Server	Subnet Mask	Max Users
CustomerA_VLAN10	192.168.1.10	192.168.1.1	192.168.7.20	255.255.255.0	50
CustomerA_VLAN20	192.168.2.10	192.168.2.1	192.168.7.20	255.255.255.0	50
CustomerA_VLAN30	192.168.3.10	192.168.3.1	192.168.7.20	255.255.255.0	50
CustomerB_VLAN10	192.168.4.10	192.168.4.1	192.168.7.20	255.255.255.0	50

CustomerB_VLAN20	192.168.5.10	192.168.5.1	192.168.7.20	255.255.255.0	50
CustomerB_VLAN30	192.168.6.10	192.168.6.1	192.168.7.20	255.255.255.0	50

DHCP Helper-Address on R2 and R3 (applied to each sub-interface):

R2(config)#interface gigabitEthernet 0/1.10

R2(config-if)#ip helper-address 192.168.7.10

R2(config)#interface gigabitEthernet 0/1.20

R2(config-if)#ip helper-address 192.168.7.10

R2(config)#interface gigabitEthernet 0/1.30

R2(config-if)#ip helper-address 192.168.7.10

R3(config)#interface gigabitEthernet 0/1.10

R3(config-if)#ip helper-address 192.168.7.10

R3(config-if)#exit

R3(config)#interface gigabitEthernet 0/1.20

R3(config-if)#ip helper-address 192.168.7.10

R3(config-if)#exit

R3(config)#interface gigabitEthernet 0/1.30

R3(config-if)#ip helper-address 192.168.7.10

R3(config-if)#exit

R3(config)#exit

R3#copy running-config startup-config

Step 10: DNS Server Configuration

Assign static IP to DNS-Server: IP 192.168.7.20, Subnet 255.255.255.0, Gateway 192.168.7.1, DNS 192.168.7.20 (self). Enable DNS service ON.

DNS Records:

Hostname / Domain Name	Type	IP Address
www.customerA.com	A Record	192.168.1.1
www.customerB.com	A Record	192.168.4.1
www.isp.com	A Record	192.168.7.1
dhcp.isp.com	A Record	192.168.7.10
dns.isp.com	A Record	192.168.7.20

Verification:

Part A: Router Verification Commands

Interface Status – R1:

R1#show ip interface brief

```
Interface          IP-Address      OK? Method Status   Protocol
GigabitEthernet0/0 192.168.0.1     YES manual up       up
GigabitEthernet0/1 192.168.0.5     YES manual up       up
GigabitEthernet0/2 192.168.7.1     YES manual up       up
Serial0/0/0        203.0.113.1     YES manual up       up
```

OSPF Neighbor Table – R1:

R1#show ip ospf neighbor

```
Neighbor ID  Pri  State   Dead Time  Address        Interface
2.2.2.2       1   FULL/-  00:00:38   192.168.0.2    GigabitEthernet0/0
3.3.3.3       1   FULL/-  00:00:37   192.168.0.6    GigabitEthernet0/1
```

Routing Table – R1 (key entries):

R1#show ip route

O 192.168.1.0/22 [110/2] via 192.168.0.2 (Customer A Summary)

O 192.168.4.0/22 [110/2] via 192.168.0.6 (Customer B Summary)

C 192.168.0.0/30 is directly connected, GigabitEthernet0/0

C 192.168.0.4/30 is directly connected, GigabitEthernet0/1

C 192.168.7.0/24 is directly connected, GigabitEthernet0/2

S* 0.0.0.0/0 is directly connected, Serial0/0/0

Part B: Switch Verification Commands

SW1#show vlan brief

VLAN Name Status Ports

1 default active Gi0/2, Fa0/4-24

10 CustomerA_Dept1 active Fa0/1

20 CustomerA_Dept2 active Fa0/2

30 CustomerA_Dept3 active Fa0/3

SW1#show interfaces trunk

Port Mode Encapsulation Status Native vlan

Gi0/1 on 802.1q trunking 1

Port Vlans allowed on trunk

Gi0/1 10,20,30

Part C: PC DHCP Verification

PC	VLAN	Assigned IP	Subnet Mask	Gateway	DNS Server
PC1	VLAN 10	192.168.1.10	255.255.255.0	192.168.1.1	192.168.7.20

PC2	VLAN 20	192.168.2.10	255.255.255.0	192.168.2.1	192.168.7.20
PC3	VLAN 30	192.168.3.10	255.255.255.0	192.168.3.1	192.168.7.20
PC4	VLAN 10	192.168.4.10	255.255.255.0	192.168.4.1	192.168.7.20
PC5	VLAN 20	192.168.5.10	255.255.255.0	192.168.5.1	192.168.7.20
PC6	VLAN 30	192.168.6.10	255.255.255.0	192.168.6.1	192.168.7.20

Part D: Connectivity Tests from PC1

C:\> ipconfig -- Verify DHCP assignment

C:\> ping 192.168.1.1 -- Ping default gateway

C:\> ping 192.168.7.10 -- Ping DHCP Server

C:\> ping 192.168.7.20 -- Ping DNS Server

C:\> ping 192.168.4.10 -- Ping PC4 (cross-customer)

C:\> ping www.customerA.com -- DNS resolution test

C:\> tracert 192.168.4.10 -- Trace route to Customer B

Expected traceroute output:

1 <1ms 192.168.1.1 (R2 VLAN 10 gateway)

2 1ms 192.168.0.1 (R1 core)

3 1ms 192.168.0.6 (R3 link)

4 2ms 192.168.4.10 (PC4)

Troubleshooting Guide:

Problem	Symptoms	Solutions
Interfaces Down/Down	Red triangles; admin down in show brief	Apply no shutdown; check cabling; verify correct cable type
OSPF Neighbors Not Forming	No neighbors in show ip ospf neighbor	Verify network statements; check area numbers match (area 0); ping between router interfaces
VLANs Not Working	PCs on same VLAN cannot communicate	Run show vlan brief; verify access port assignments; check trunk allows correct VLANs
DHCP Not Assigning IPs	PCs stuck with 169.254.x.x address	Verify DHCP ON; check ip helper-address on each sub-interface; ping 192.168.7.10 from router
DNS Not Resolving	Ping by IP works, name fails	Verify DNS ON; check A records; confirm PCs received DNS 192.168.7.20 via DHCP
No Internet Access	Internal works, external fails	Check default route on R1; verify default-information originate; confirm cloud interface is ON
Summary Routes Not Showing	Individual /24 routes on R1	Verify area 0 range command on R2/R3; check show ip route on R1 for /22 entries

Verification Checklist:

SN	Verification Task	Expected Result	Status
1	All router interfaces UP/UP	show ip interface brief – all green	✓
2	OSPF neighbors R1-R2 and R1-R3	FULL state in show ip ospf neighbor	✓

3	Summary routes on R1	O 192.168.1.0/22 and O 192.168.4.0/22	✓
4	Default route propagated	O*E2 0.0.0.0/0 on R2 and R3	✓
5	VLANs created on SW1 and SW2	show vlan brief – VLANs 10,20,30 active	✓
6	Trunk ports active	show interfaces trunk – trunking status	✓
7	All PCs receive DHCP IP	PC Desktop → IP Config shows dynamic IP	✓
8	PCs ping their gateways	Reply from gateway IP	✓
9	Inter-VLAN communication	PC1 pings PC2 and PC3	✓
10	Cross-router communication	PC1 pings PC4 (different customer)	✓
11	DNS resolution working	ping www.customerA.com returns reply	✓
12	All servers reachable	Ping 192.168.7.10 and 192.168.7.20	✓

Output on verification,testing,pinging

Router Verification Commands

1. Verify Interface Status on R1:

R1#show ip interface brief

```
[OK]
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.0.1     YES manual up                    up
GigabitEthernet0/1     192.168.0.5     YES manual up                    up
GigabitEthernet0/2     192.168.7.1     YES manual up                    up
Serial0/0/0            203.0.113.1     YES manual up                    down
Serial0/0/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

2. Verify Interface Status on R2

R2#show ip interface brief

```
R2>enable
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.0.2     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  up                    up
GigabitEthernet0/1.10  192.168.1.1     YES manual up                    up
GigabitEthernet0/1.20  192.168.2.1     YES manual up                    up
GigabitEthernet0/1.30  192.168.3.1     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R2#
```

3. Verify Interface Status on R3

R3#show ip interface brief

```
R3>enable
R3#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.0.6     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  up                    up
GigabitEthernet0/1.10  192.168.4.1     YES manual up                    up
GigabitEthernet0/1.20  192.168.5.1     YES manual up                    up
GigabitEthernet0/1.30  192.168.6.1     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R3#
```

4. Verify OSPF Neighbors on R1

R1#show ip ospf neighbor

```
R1#
R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           1   FULL/DR         00:00:35    192.168.0.6     GigabitEthernet0/1
2.2.2.2           1   FULL/DR         00:00:35    192.168.0.2     GigabitEthernet0/0
R1#
```

5. Verify OSPF Neighbors on R2

R2#show ip ospf neighbor

```
R2#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/BDR        00:00:33    192.168.0.1     GigabitEthernet0/0
R2#
```

6. Verify OSPF Neighbors on R3:

R3#show ip ospf neighbor

```
R3#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/BDR        00:00:33    192.168.0.5     GigabitEthernet0/0
R3#
```

7. Verify Routing Table on R1:

R1#show ip route

```
R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interfac
3.3.3.3           1   FULL/DR         00:00:35    192.168.0.6     GigabitE
2.2.2.2           1   FULL/DR         00:00:35    192.168.0.2     GigabitE
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.0.0/24 is variably subnetted, 4 subnets, 2 masks
C       192.168.0.0/30 is directly connected, GigabitEthernet0/0
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0
C       192.168.0.4/30 is directly connected, GigabitEthernet0/1
L       192.168.0.5/32 is directly connected, GigabitEthernet0/1
O    192.168.1.0/24 [110/2] via 192.168.0.2, 00:26:11, GigabitEthernet0/0
O    192.168.2.0/24 [110/2] via 192.168.0.2, 00:26:11, GigabitEthernet0/0
O    192.168.3.0/24 [110/2] via 192.168.0.2, 00:26:11, GigabitEthernet0/0
O    192.168.4.0/24 [110/2] via 192.168.0.6, 01:15:58, GigabitEthernet0/1
O    192.168.5.0/24 [110/2] via 192.168.0.6, 01:15:58, GigabitEthernet0/1
O    192.168.6.0/24 [110/2] via 192.168.0.6, 01:15:58, GigabitEthernet0/1
     192.168.7.0/24 is variably subnetted, 2 subnets, 2 masks
 --More--
```

8. Verify Routing Table on R2:

R2#show ip route

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

O    192.168.0.0/22 is a summary, 00:00:00, Null0
     192.168.0.0/24 is variably subnetted, 3 subnets, 2 masks
C       192.168.0.0/30 is directly connected, GigabitEthernet0/0
L       192.168.0.2/32 is directly connected, GigabitEthernet0/0
O       192.168.0.4/30 [110/2] via 192.168.0.1, 00:27:49, GigabitEthernet0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/1.10
L       192.168.1.1/32 is directly connected, GigabitEthernet0/1.10
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/1.20
L       192.168.2.1/32 is directly connected, GigabitEthernet0/1.20
     192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
 --More--
```

9. Verify Routing Table on R3:

R3#show ip route

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.0.0/24 is variably subnetted, 3 subnets, 2 masks
O       192.168.0.0/30 [110/2] via 192.168.0.5, 00:28:41, GigabitEthernet0/0
C       192.168.0.4/30 is directly connected, GigabitEthernet0/0
L       192.168.0.6/32 is directly connected, GigabitEthernet0/0
O    192.168.1.0/24 [110/3] via 192.168.0.5, 00:28:41, GigabitEthernet0/0
O    192.168.2.0/24 [110/3] via 192.168.0.5, 00:28:41, GigabitEthernet0/0
O    192.168.3.0/24 [110/3] via 192.168.0.5, 00:28:41, GigabitEthernet0/0
     192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.4.0/24 is directly connected, GigabitEthernet0/1.10
L       192.168.4.1/32 is directly connected, GigabitEthernet0/1.10
     192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.5.0/24 is directly connected, GigabitEthernet0/1.20
 --More--
```

10. Verify OSPF Configuration:

R1#show ip protocols

```
R1#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.0.0 0.0.0.3 area 0
    192.168.0.4 0.0.0.3 area 0
    192.168.7.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:13:08
    2.2.2.2              110      00:29:57
    3.3.3.3              110      00:29:52
  Distance: (default is 110)

R1#
```

Switch Verification Commands

11. Verify VLANs on SW1:

SW1#show vlan brief

```
SW1>enable
SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/2
10   CustomerA_Dept1                  active    Fa0/1
20   CustomerA_Dept2                  active    Fa0/2
30   CustomerA_Dept3                  active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW1#
```

12. Verify VLANs on SW2:

SW2#show vlan brief

```
SW2>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/2
10   CustomerB_Dept1                  active    Fa0/1
20   CustomerB_Dept2                  active    Fa0/2
30   CustomerB_Dept3                  active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW2>
```

13. Verify Trunk on SW1:

SW1#show interfaces trunk

```
SW1#show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20,30

Port        Vlans allowed and active in management domain
Gig0/1      10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,30

SW1#
```

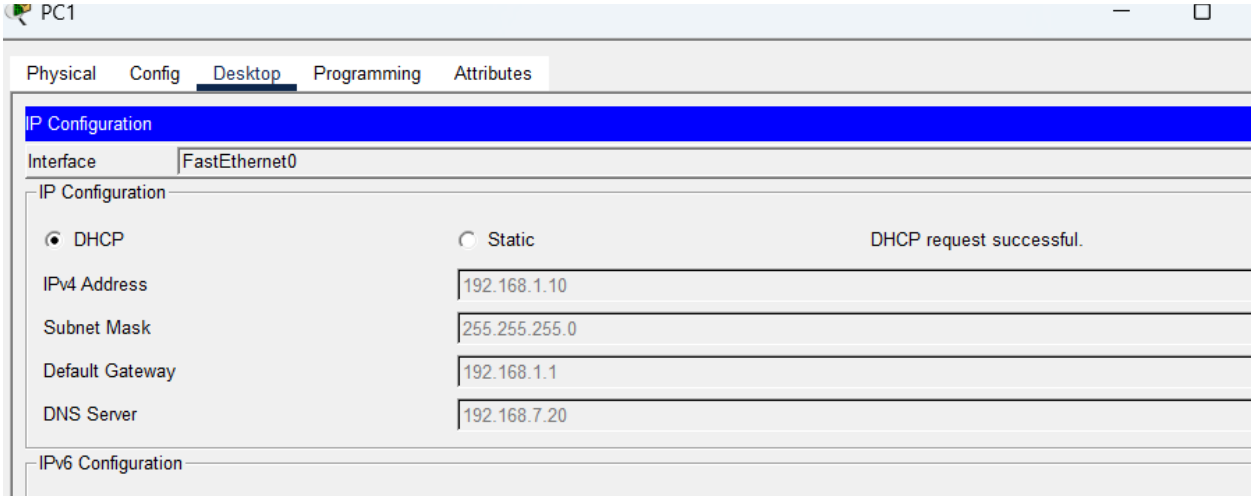
14. Verify Trunk on SW2:

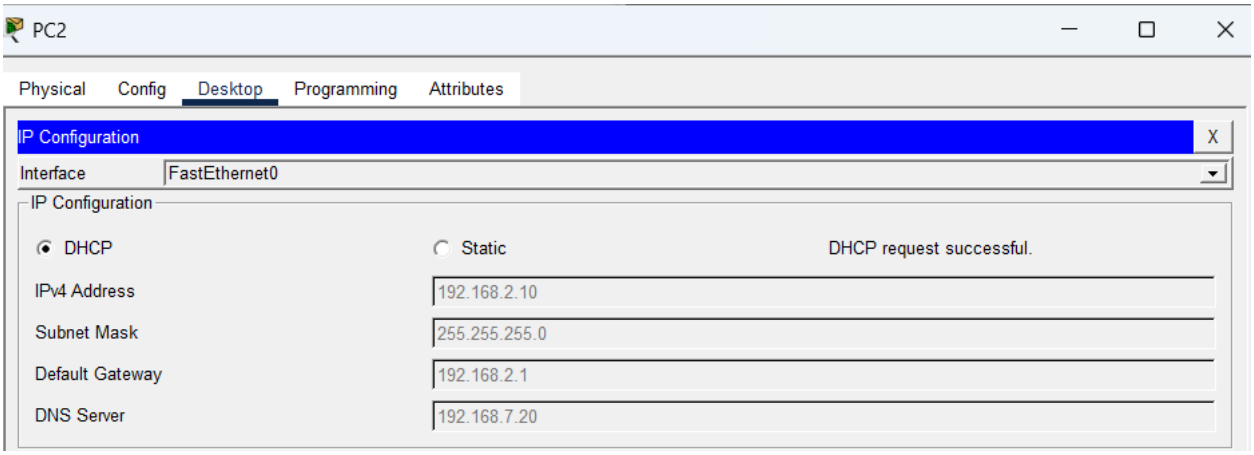
SW2#show interfaces trunk

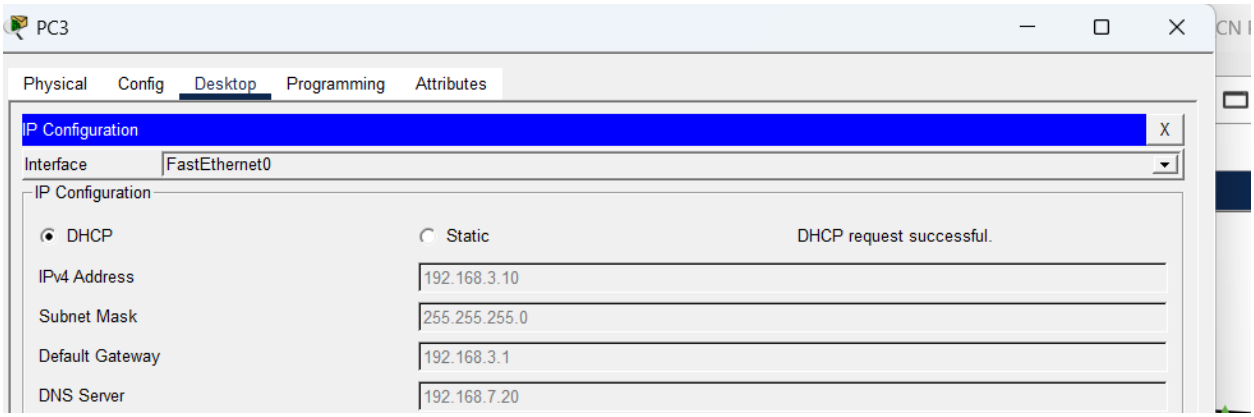
```
SW2>show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20,30

Port        Vlans allowed and active in management domain
Gig0/1      10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,30

SW2>
```

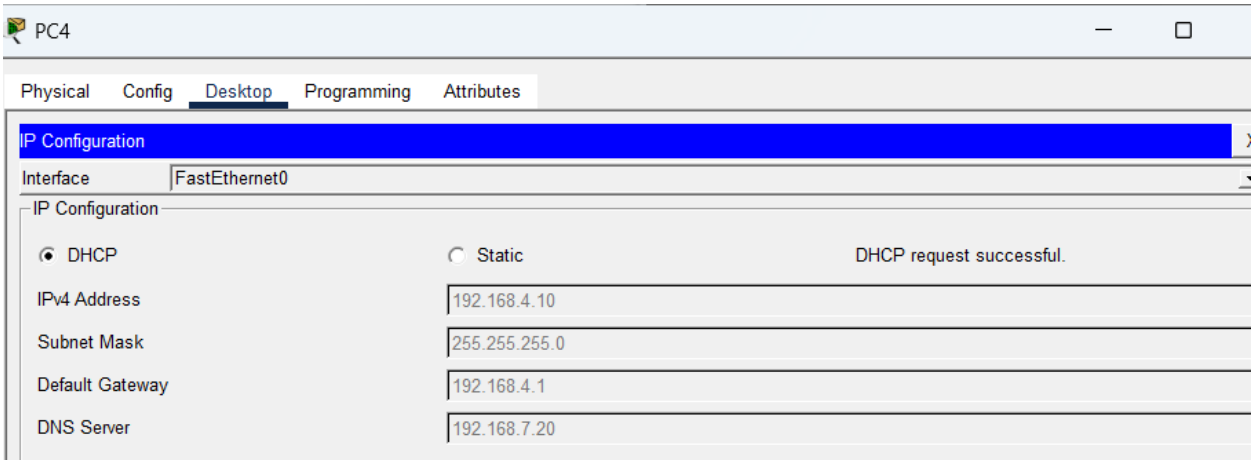
End Device Testing

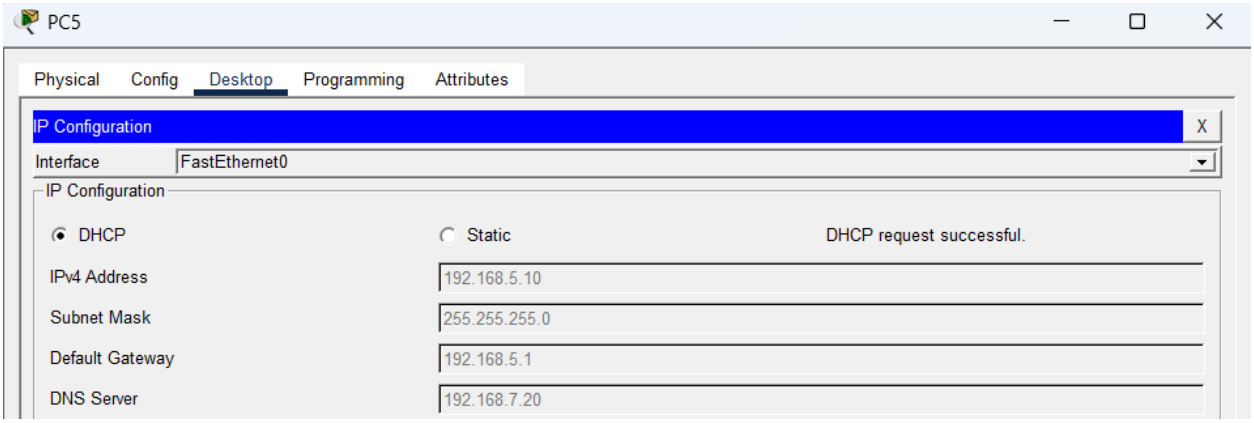
15. Configure PCs to Use DHCP:

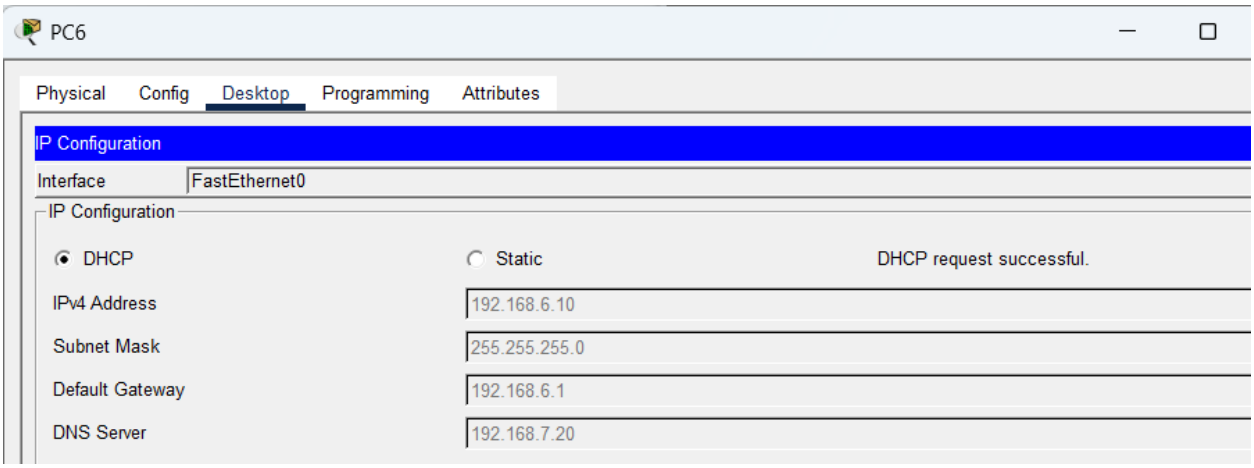












16. Pinging from PC

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ip config
Invalid Command.

C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::290:21FF:FE77:AA86
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.10
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.1.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

17. Ping Test from PC1 to DHCP Server:

C:\> ping 192.168.7.10

```
:\>ping 192.168.7.10

inging 192.168.7.10 with 32 bytes of data:

eply from 192.168.7.10: bytes=32 time<1ms TTL=126
eply from 192.168.7.10: bytes=32 time<1ms TTL=126
eply from 192.168.7.10: bytes=32 time<1ms TTL=126
eply from 192.168.7.10: bytes=32 time<1ms TTL=126

ing statistics for 192.168.7.10:
   Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
pproximate round trip times in milli-seconds:
   Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

18. Ping Test from PC1 to PC4 (Different Customer):

C:\> ping 192.168.4.10

```
C:\>ping 192.168.4.10

Pinging 192.168.4.10 with 32 bytes of data:

Reply from 192.168.4.10: bytes=32 time<1ms TTL=125
Reply from 192.168.4.10: bytes=32 time<1ms TTL=125
Reply from 192.168.4.10: bytes=32 time<1ms TTL=125
Reply from 192.168.4.10: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.4.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

19. DNS Test from PC1:

C:\> ping www.customerA.com

```
C:\>ping www.CustomerA.com

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=10ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

20. Traceroute Test from PC1 to PC4:

C:\> tracert 192.168.4.10

```
C:\>tracert 192.168.4.10

Tracing route to 192.168.4.10 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.1.1
  2   0 ms      0 ms      0 ms      192.168.0.1
  3   0 ms      0 ms      0 ms      192.168.0.6
  4   0 ms      0 ms      0 ms      192.168.4.10

Trace complete.
```

Result:

The ISP network was successfully designed and implemented in Cisco Packet Tracer. All three routers formed OSPF adjacencies in the FULL state. Route summarization reduced Customer A and Customer B routes to single /22 summary entries on R1. VLANs were created and trunked correctly on SW1 and SW2. All six PCs automatically received correct IP addresses, gateways, and DNS server settings via DHCP. DNS name resolution worked successfully for all configured domains. Default route propagation via OSPF ensured all customer PCs had a path to the Internet through R1.

Discussion:

This lab demonstrated the integration of multiple enterprise networking technologies into a functional ISP environment. OSPF was chosen over static routing because it automatically adapts to topology changes and scales well, while the Router-on-a-Stick technique proved effective for providing inter-VLAN routing without requiring a Layer 3 switch, using 802.1Q sub-interfaces on a single physical port. Supernetting (route aggregation) significantly reduced the number of OSPF LSAs flooded through the backbone — by summarizing Customer A's three /24 networks into one /22 and similarly for Customer B, R1's routing table became more concise and efficient. The ip helper-address command was equally critical for DHCP to function across routed boundaries, as without it, DHCP broadcast packets from PCs would be discarded at the router interface, but by configuring the helper address on each sub-interface, DHCP requests were forwarded as unicast to the centralized DHCP server. Together, OSPF, VLANs, DHCP, and DNS formed a complete, scalable ISP service model demonstrating real-world network design principles.

Conclusion:

The ISP Network Design lab was successfully completed. A functional three-router ISP topology was built using OSPF for dynamic routing, VLANs for customer segmentation, DHCP for automatic address assignment, DNS for name resolution, and a default route for Internet connectivity. Route aggregation reduced backbone routing overhead, and all verification tests confirmed end-to-end connectivity across all customer sites. This lab highlights the importance of combining multiple networking protocols and techniques to deliver efficient, scalable, and manageable ISP-grade network infrastructure.